

Industrial Dyeing Machinery Constraints & Operational Guide

1. Overview of Machine Constraints in Textile Dyeing

In industrial textile wet processing, machine constraints define the physical, thermodynamic, and mechanical boundaries within which a dyeing process can be executed safely, efficiently, and with repeatable quality. Key parameters affected by machine design include liquor-to-material ratio, thermal heating/cooling capabilities, pressure limits, pump flow rate, liquor circulation frequency, nozzle dynamics, and mechanical tension.

2. Key Machine Constraint Parameters

2.1 Liquor-to-Material Ratio (MLR / Liquor Ratio):

- Conventional Winch: 1:15 to 1:30
- Conventional Jigger: 1:3 to 1:5
- HTHP Soft-Flow / Jet Machine: 1:5 to 1:8
- Airflow Dyeing Machine: 1:3 to 1:5
- HTHP Package Dyeing Machine: 1:6 to 1:10

2.2 Temperature & Pressure Boundaries:

- Atmospheric Machines (Winch, Jigger): Max 95°C – 98°C at sea level.
- HTHP Machines (Jet, Package, Beam): Max design pressure 3.5 – 4.5 bar, operating temp 130°C – 140°C.

2.3 Thermal Ramping (Heating & Cooling Rates):

- Heating Gradient: 0.5°C/min to 3.0°C/min (1.0°C – 1.5°C/min through critical strike zone).
- Cooling Gradient: 1.5°C/min to 2.5°C/min (controlled cooling to prevent oligomers and thermal shock).

2.4 Pump Capacity & Flow Rate:

- Circulation Rate: 35 – 50 L/kg/min for spun yarns; 25 – 35 L/kg/min for filaments.
- Differential Pressure (ΔP): 0.8 bar – 1.5 bar.

2.5 Fabric Velocity & Nozzle Pressure:

- Recommended Turn Time: 0.8 to 1.5 minutes (48 – 90 seconds).
- Fabric Speed: 150 to 450 m/min; Nozzle Pressure: 0.2 to 1.2 bar.

3. Machinery Comparison Matrix

Machine Type	MLR	Temp (°C)	Pressure	Substrate	Primary Constraint
HTHP Jet	1:5 - 1:8	Up to 140°C	3.5-4.0 bar	Knits, PES, Blends	Nozzle damage, turn time
Air-Jet	1:3 - 1:5	Up to 140°C	3.5-4.0 bar	Delicate knits, Lycra	Blower power, foaming
HTHP Package	1:6 - 1:10	Up to 135°C	4.0 bar	Cotton/PES Yarns	Package density (±5%)
Atm. Jigger	1:3 - 1:5	Up to 98°C	Atmospheric	Woven (Cotton/Viscose)	Shade variation, tension
Pad-Steam	Pad 65%	102°C	Atmospheric	Woven long runs	Pick-up consistency
Beam Dyeing	1:8 - 1:12	Up to 130°C	3.5 bar	Warp knits, Tulle	Moiré effect, hardness